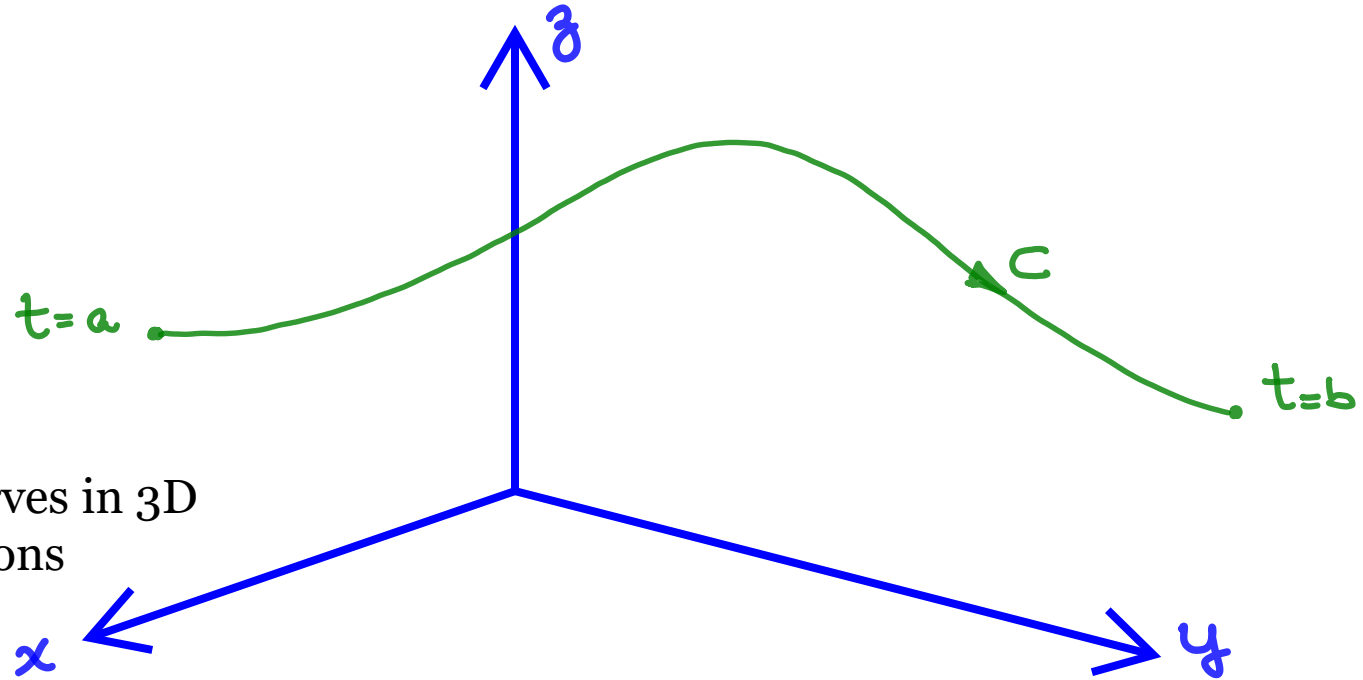


Parametrization of Curves in 3D Space and its Applications



Outline:

1. Parametrizing curves in 3D
2. Physical applications



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Curves formed by the intersection of surfaces

Problem:

Parametrize the curve of intersection of the plane:

$$x + 2y + 4z = 4$$

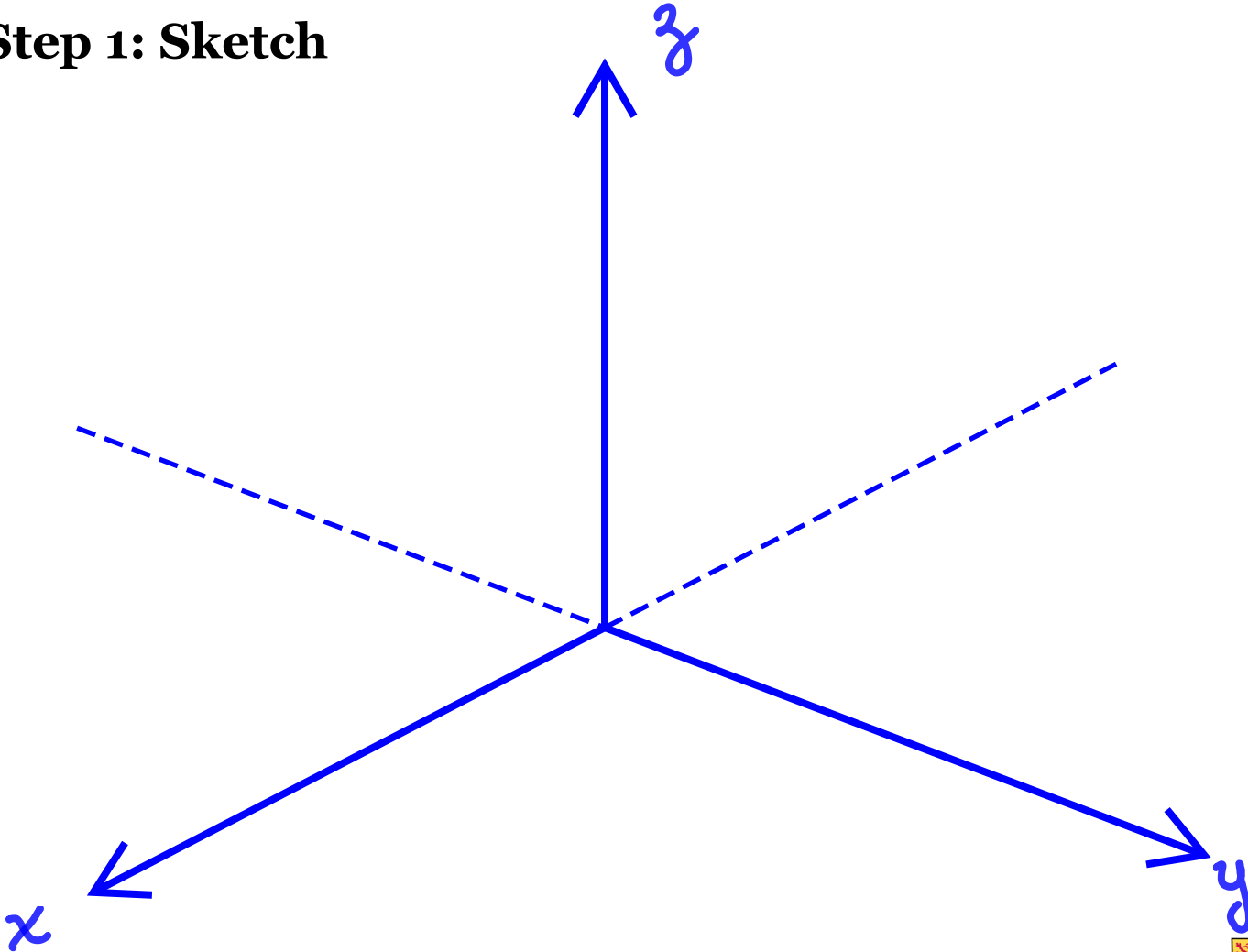
And the elliptical cylinder:

$$x^2 + 4y^2 = 4$$

Directed so that “y’ increases when x is positive.

Curves formed by the intersection of surfaces

Step 1: Sketch



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Curves formed by the intersection of surfaces

Step 2: Look for a convenient way to parametrize

Curves formed by the intersection of surfaces

Curves formed by the intersection of surfaces

Step 3: Check direction

Curves formed by the intersection of surfaces

Problem:

Parametrize the curve of intersection of the surfaces:

$$z = \sqrt{x^2 + y^2}$$

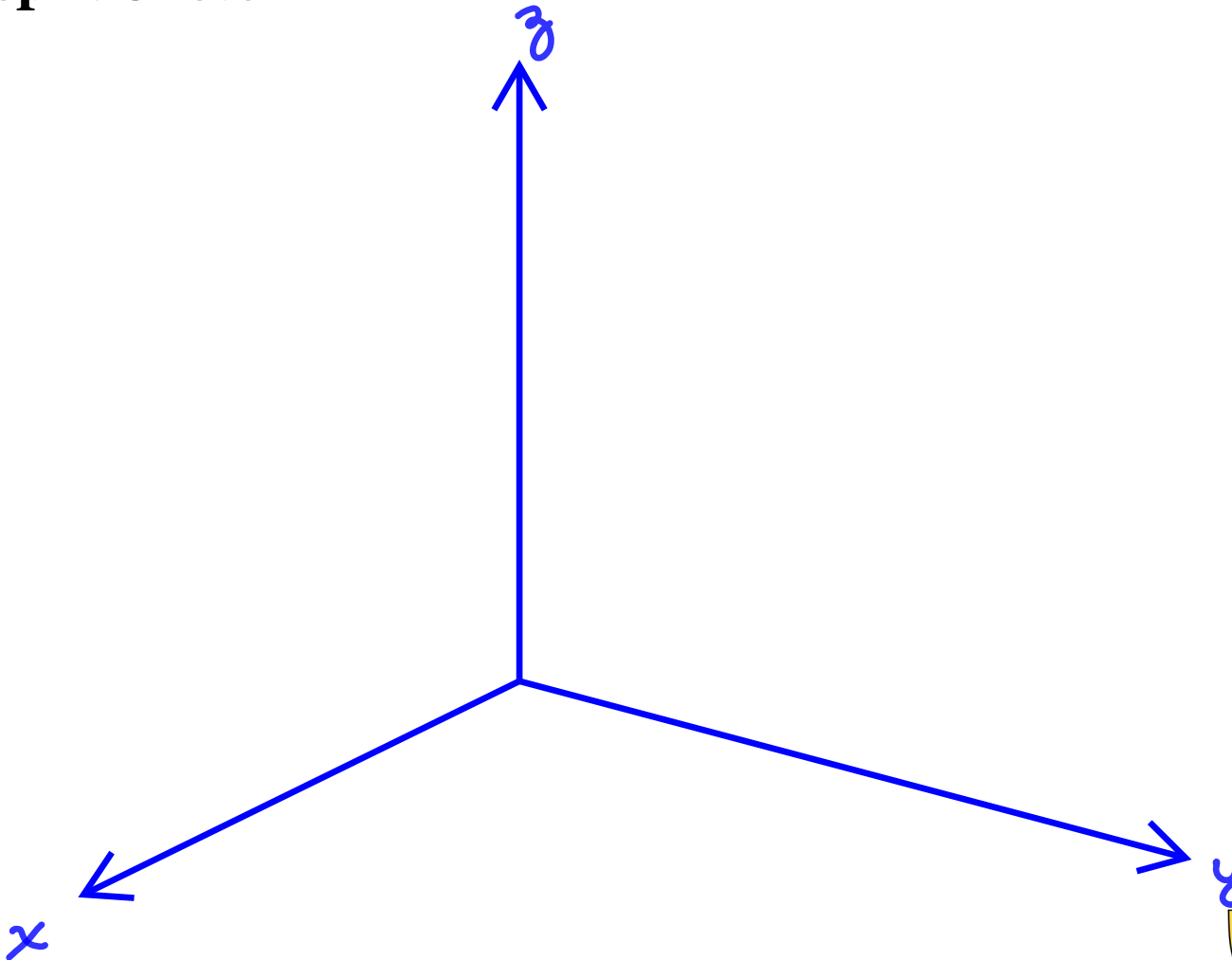
and

$$y = x^2$$

Directed so that “y’ decreases in the first octant.

Curves formed by the intersection of surfaces

Step 1: Sketch



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Curves formed by the intersection of surfaces

Step 2: Look for a convenient way to parametrize

Step 3: Check direction

Curves formed by the intersection of surfaces

Parametrizing Curves

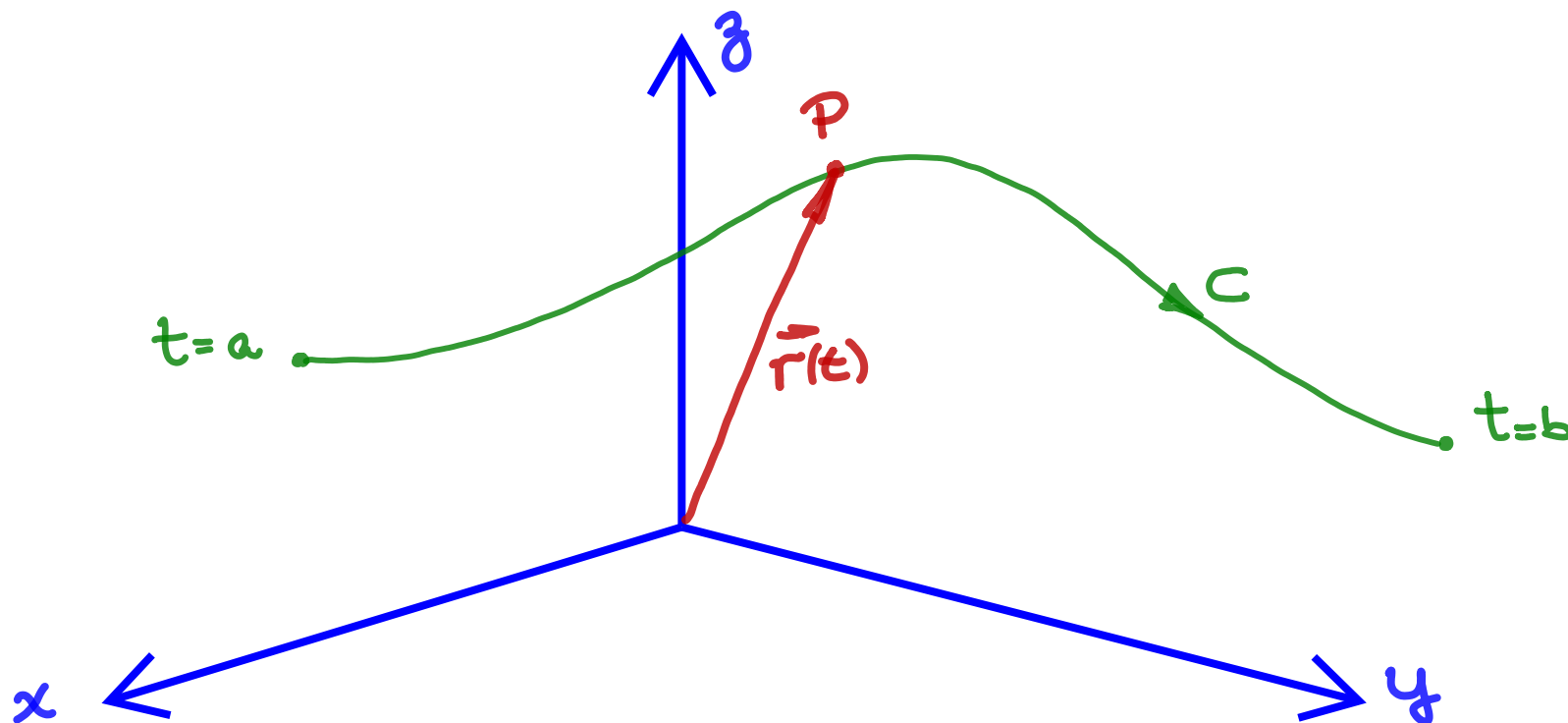
- To parametrize a curve formed by the intersection of surfaces
 1. Draw a 3D sketch (a.k.a. graph) of the curve of intersection
 2. After observing the surface equations, determine a parametric equation for each variable and determine the parameter range
 3. Check the direction of the curve (plot 3 points minimum)
 4. If direction is not correct, determine new parametric equations (usually a change of sign is a first good try)
- There exists infinite parametrizations for the same curve

With some more practice you will “master the art” of parametrization

Physical Applications

Let us consider the general form of a vector function describing a curve C :

$$C : \vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}, \quad a \leq t \leq b$$

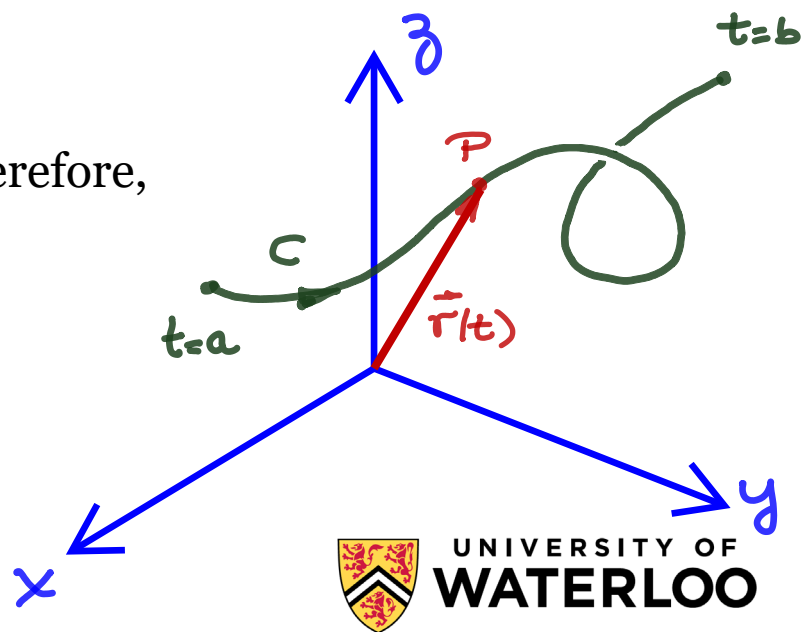


We can see the usefulness of vector functions in the analysis of motion of a particle!

Differentiation of Vector Functions

$\vec{r}(t)$ is differentiable if it is differentiable at every point of its domain, and

The rate of change of $\vec{r}(t)$ as a function of t is therefore,



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Summary

- We learned how to parametrize a curve formed by the intersection of surfaces
- We defined the rate of change of $\vec{r}(t)$ as a function of t
- **Next lecture:**

How can we apply this knowledge?



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Thank you